

name of these living benefit guarantees, I can't help but wonder why Wall Street's investment bankers charge so much more for the same type of derivative security (essentially long-term put options) when they are purchased on a stand-alone basis. To review once more, put options are contracts that offer the purchaser a right but not the obligation to sell an asset at a predetermined "strike" price. And when I obtained some pricey quotes for buying stand-alone put options to protect a hypothetical retiree's lifetime income, I first thought it was the derivatives dealers and option market makers that were overcharging. To illustrate these high costs, in Table 9.5 I include the approximate cost of protecting \$500,000 invested in a hypothetical fund (that exactly tracks the performance of the SP500) for one month, one year, and two years. Based on the cost of a one-month put option, it would cost approximately 2.73% or \$13,650!

**TABLE 9.5 Cost of Protection in Capital Markets
(SP500 Spot Price =\$1,440)**

Option Expiration	Upfront Cost	Annualized Cost
1 Month	2.73%	—
1 Year	9.08%	908 b.p.
2 Years	11.97%	599 b.p.

Source: Chicago Board Options Exchange put price quotes as of 11/22/07 and IFID Centre calculations; costs are an upper bound.

But, after some careful analysis, the same mathematical models that told us a decade ago that basic death benefit guarantees were overpriced were now telling us that many living benefits, for the most part, were underpriced.

As an example, I recently conducted a follow-up study with a colleague of mine at York University demonstrating that the basic GMWB rider, which charges an extra 30 to 50 b.p. in practice, might actually cost between 75 and 160 b.p. to hedge in the capital markets. And, that number does not even include any insurance company profit